

UML Use-Case Diagram

Lab 1 – Requirements Engineering & UML Use-Case Modelling

Problem Statement #44 – Database Query Performance Profiler

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Actors:

- Database Administrator – main user, monitors performance and acts on recommendations
- Backend Lead – receives weekly digest emails and checks the dashboard
- Scheduled Job Trigger – automated cron job that fires every Monday at 8 AM to generate the digest

Use Cases:

- UC-01: Ingest Slow Query Logs
- UC-02: Parse EXPLAIN Plans
- UC-03: Identify Missing Indexes
- UC-04: Generate Weekly Digest
- UC-05: View Dashboard

Relationships:

- UC-04 <> UC-02 – generating a digest always requires parsing the execution plans
- UC-02 <> UC-03 – analysing a plan always involves checking for missing indexes
- UC-05 <> UC-03 – the dashboard can optionally trigger an index check, but not always

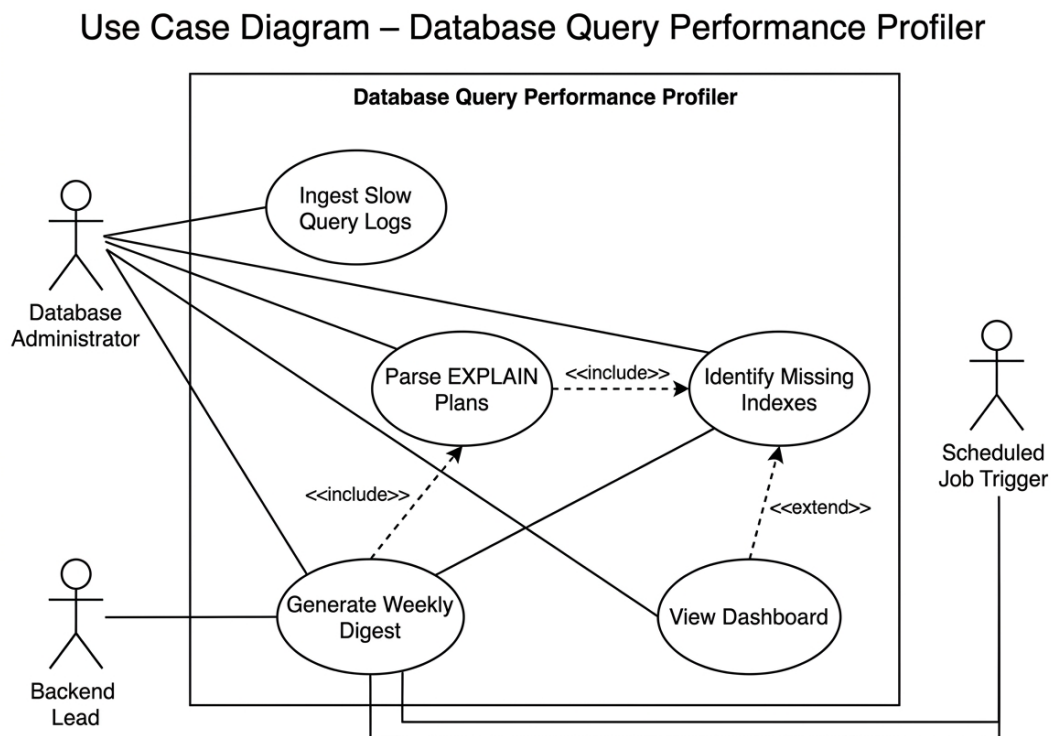


Figure 1 – Use-Case Diagram (draw.io)